Transformation of Rational Functions

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1. The six basic transformations $(a > 0)$

New graph	Transformation	Point $(x, y) \to$
$y = f(x) + b$	vertical translation b units up	$(x,\ y + b)$
$y = f(x - a)$	horizontal translation a units **right**	$(x + a,\ y)$
$y = p\,f(x)$	vertical stretch, scale factor p	$(x,\ py)$
$y = f(qx)$	horizontal stretch, scale factor $\frac{1}{q}$	$\left(\frac{x}{q},\ y\right)$
$y = -f(x)$	reflection in the x-axis	$(x,\ -y)$
$y = f(-x)$	reflection in the y-axis	$(-x,\ y)$

Golden rule: outside the bracket acts on y exactly as written; **inside** the bracket acts on x the **opposite** way ($+$ moves left, $\times$ shrinks).

Translation by vector $\begin{pmatrix} h \\ k \end{pmatrix}$: $y = f(x - h) + k$.

2. Combining transformations — the order matters

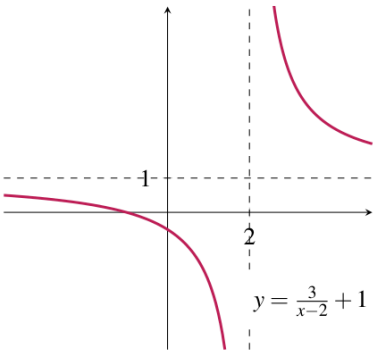
- **Vertical** (y): follow the order of operations on f: stretch / reflect first, then translate. $y = 2f(x) + 4$: stretch s.f. 2, then 4 up.
- **Horizontal** (x): undo what was done to x in reverse. $f(2x - 10)$: translate 10 right, then stretch s.f. $\frac{1}{2}$. $f(2(x - 5))$: stretch s.f. $\frac{1}{2}$, then translate 5 right (same graph!).
- Horizontal and vertical changes do not affect each other, so they can be done in either order.
- Follow one point through each step: $A(-1, 3)$ under $2f(x - 3) + 4$: $(-1, 6) \to (2, 6) \to (2, 10)$.

$y = a(x-h)^2 + k$ is $y = x^2$ stretched vertically by s.f. a, then translated by $\begin{pmatrix} h \\ k \end{pmatrix}$.

- Vertex (h,k), axis of symmetry $x = h$ (write it as an **equation**). If $a < 0$ there is also a reflection in the x-axis (maximum).
- Completing the square: $3x^2 - 12x + 11 = 3(x^2 - 4x) + 11 = 3\left[(x-2)^2 - 4\right] + 11 = 3(x-2)^2 - 1$.
- Transform the vertex directly: reflect in x-axis $(h,k) \to (h,-k)$; in y-axis $(h,k) \to (-h,k)$.

4. Rational functions $y = \dfrac{a}{x-h} + k$

This is $y = \dfrac{1}{x}$ stretched vertically by s.f. a, then translated by $\begin{pmatrix} h \\ k \end{pmatrix}$.



- Vertical asymptote $x = h$; horizontal asymptote $y = k$.
- Domain $x \neq h$; range $y \neq k$.
- Intercepts: put $x = 0$, then $y = 0$.
- $y = \dfrac{ax+b}{cx+d}$: asymptotes $x = -\dfrac{d}{c}$, $y = \dfrac{a}{c}$.
  Divide first: $\dfrac{2x+5}{x+1} = 2 + \dfrac{3}{x+1}$.

5. Exam tips

- "Describe **fully**": name the transformation **and** give the details — "vertical stretch, scale factor 3", "translation by $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$", "reflection in the x-axis".
- Domain changes with horizontal transformations; range changes with vertical ones.
- $[f(x)]^2$: square every y-value (negatives become positive); asymptote $y = c$ becomes $y = c^2$.
- Sketches: label intercepts, turning points and asymptotes (as equations).

Question 1

Source worksheet · p.5

1

No calculator **Chris · Math-AA 2.6 · Q1**

[Maximum mark: 9] ***[without GDC]***

The point A(1, 0.5) lies on the curve $y = f(x)$. Write down the coordinates of the corresponding point under the following transformations

$y = f(x) + 5$	(1, 5.5)	$y = f(x + 5)$	
$y = f(x) - 5$		$y = f(x - 5)$	
$y = 5f(x)$		$y = f(5x)$	
$y = f(x) / 5$		$y = f(x / 5)$	
$y = -f(x)$		$y = f(-x)$	

Question 2

Source worksheet · p.5

2

No calculator **Chris · Math-AA 2.6 · Q2**

[Maximum mark: 12] ***[without GDC]***

The point A(-1, 3) lies on the curve $y = f(x)$.

(a) Write down the coordinates of the corresponding point under the following transformations [9]

$y = f(x) + 3$	(-1, 6)	$y = f(x + 3)$	
$y = f(x) - 3$		$y = f(x - 3)$	
$y = 3f(x)$		$y = f(3x)$	
$y = f(x) / 3$		$y = f(x / 3)$	
$y = -f(x)$		$y = f(-x)$	

(b) Find the coordinates of the corresponding point under $y = 2f(x-3)+4$. [3]

Answer 1 Source worksheet · p.5

1 *Chris · Math-AA 2.6 · Q1*

$f(x)+5$	$(1, 5.5)$	$f(x+5)$	$(-4, 0.5)$
$f(x)-5$	$(1, -4.5)$	$f(x-5)$	$(6, 0.5)$
$5f(x)$	$(1, 2.5)$	$f(5x)$	$(0.2, 0.5)$
$f(x)/5$	$(1, 0.1)$	$f(x/5)$	$(5, 0.5)$
$-f(x)$	$(1, -0.5)$	$f(-x)$	$(-1, 0.5)$

Outside the bracket changes y; inside the bracket changes x (the opposite way).

Answer 2 Source worksheet · p.5

2 *Chris · Math-AA 2.6 · Q2*

(a)

$f(x)+3$	$(-1, 6)$	$f(x+3)$	$(-4, 3)$
$f(x)-3$	$(-1, 0)$	$f(x-3)$	$(2, 3)$
$3f(x)$	$(-1, 9)$	$f(3x)$	$(-\frac{1}{3}, 3)$
$f(x)/3$	$(-1, 1)$	$f(x/3)$	$(-3, 3)$
$-f(x)$	$(-1, -3)$	$f(-x)$	$(1, 3)$

(b) $(-1,3) \xrightarrow{2f(x)} (-1,6) \xrightarrow{2f(x-3)} (2,6) \xrightarrow{+4} \boxed{(2, 10)}$

Question 3 Source worksheet · p.7

3 No calculator **Chris · Math-AA 2.6 · Q4**

[Maximum mark: 11] ***[without GDC]***

The sequence of transformations from $y = f(x)$ to $y = 2f(3x) + 4$ is given below

$f(x)$	original
$2f(x)$	vertical stretch with s.f. 2
$2f(3x)$	horizontal stretch with s.f. 1/3 (i.e. shrink))
$2f(3x) + 4$	vertical translation 4 units up

Describe similarly the sequence of transformations, in a correct order, for the following functions

(a) $-f(x-2)+5$ [3]

$f(x)$	original

(b) $-3f(x+2)-1$ [4]

$f(x)$	original

(c) $f(2x-10)$ [2]

$f(x)$	original

(d) $f(2(x-5))$ [2]

$f(x)$	original

Answer 3

Source worksheet · p.7

3 *Chris · Math-AA 2.6 · Q4*

(a) $-f(x)$: reflection in the x-axis; $-f(x-2)$: horizontal translation 2 units right; $-f(x-2)+5$: vertical translation 5 units up.

(b) $-f(x)$: reflection in the x-axis; $-3f(x)$: vertical stretch s.f. 3; $-3f(x+2)$: horizontal translation 2 units left; $-3f(x+2)-1$: vertical translation 1 unit down.

(c) $f(x-10)$: horizontal translation 10 units right; then $f(2x-10)$: horizontal stretch s.f. $\frac{1}{2}$.

(d) $f(2x)$: horizontal stretch s.f. $\frac{1}{2}$; then $f(2(x-5))$: horizontal translation 5 units right.

Note (c) and (d) are the same function, $f(2x-10)=f(2(x-5))$, reached in different orders.

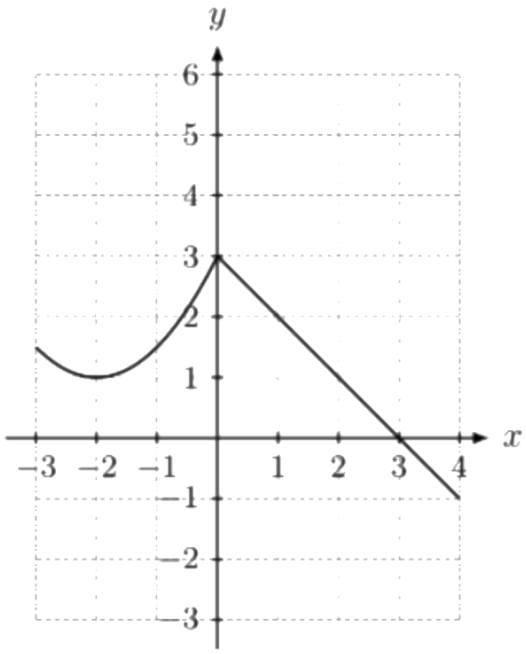
Question 4

Source worksheet · p.9

4 No calculator **RV · Revision Village Transformations · Q1**

[Maximum mark: 5]

The graph of $y = f(x)$ for $-3 \leq x \leq 4$ is shown in the following diagram.



(a) Write down the value of $f(2)$. [1]

Let $g(x) = 2f(x) - 1$ for $-3 \leq x \leq 4$.

(b) On the axes above, sketch the graph of g. [2]

(c) Hence determine the value of $(g \circ f)(2)$. [1]

(d) Hence solve the equation $(f \circ g)(x) = 0$ when $x > 0$. [1]

Answer 4

Source worksheet · p.9

4 *RV · Revision Village Transformations · Q1*

(a) $\boxed{f(2) = 1}$

(b) $g(x) = 2f(x) - 1$: vertical stretch s.f. 2, then 1 unit down. Key points: $(-3, 1.5) \to (-3, 2)$, $(-2, 1) \to (-2, 1)$, $(0, 3) \to (0, 5)$, $(3, 0) \to (3, -1)$, $(4, -1) \to (4, -3)$.

(c) $(g \circ f)(2) = g(1) = 2f(1) - 1 = 2(2) - 1 = \boxed{3}$

(d) $f(g(x)) = 0 \Rightarrow g(x) = 3$ (the only zero of f is $x = 3$) $\Rightarrow 2f(x) - 1 = 3 \Rightarrow f(x) = 2$. For $x > 0$: $3 - x = 2$, $\boxed{x = 1}$

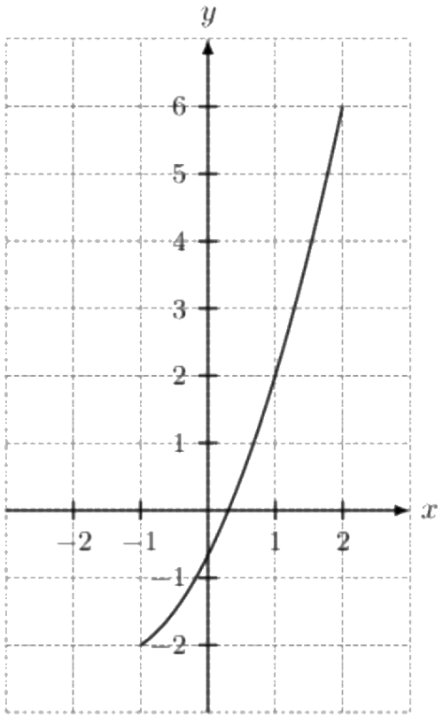
Question 5

Source worksheet · p.11

5 No calculator **RV · Revision Village Transformations · Q5**

[Maximum mark: 6]

The following diagram shows the graph of $y = f(x)$, for $-1 \leq x \leq 2$.



(a) Write down the value of:

(i) $f(1)$;

(ii) $f^{-1}(-2)$. [2]

(b) Find $(f \circ f)(1)$. [2]

(c) Sketch the graph of $y = f(-x)$ on the same grid above. [2]

Answer 5

Source worksheet · p.11

5 *RV · Revision Village Transformations · Q5*

(a)(i) $\boxed{f(1)=2}$ (ii) $f(-1)=-2$, so $\boxed{f^{-1}(-2)=-1}$

(b) $(f\circ f)(1)=f(2)=\boxed{6}$

(c) $y=f(-x)$ is the reflection of the graph in the y-axis: $(x,y)\to(-x,y)$.

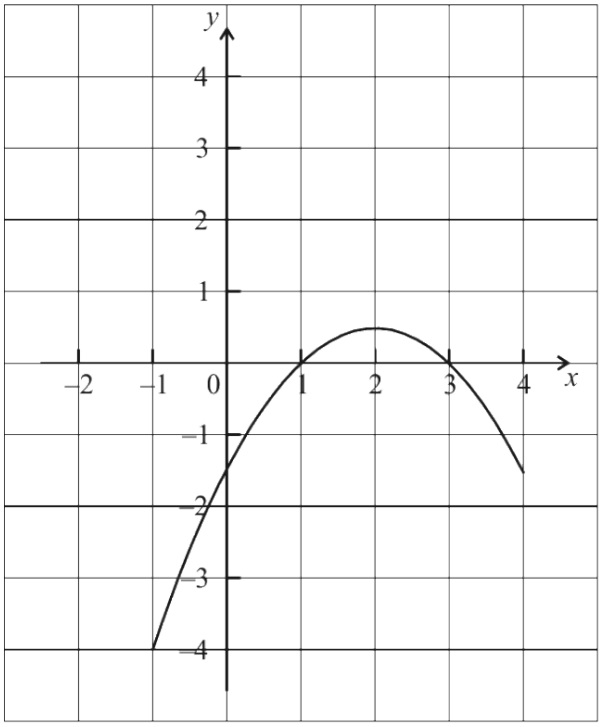
The new curve runs from $(1,-2)$ through $(0,-0.7)$, $(-1,2)$ up to $(-2,6)$.

Question 6 Source worksheet · p.13

6 No calculator **Chris · Math-AA 2.6 · Q11**

[Maximum mark: 10] ***[without GDC]***

The graph of a function f is shown in the diagram below.



(a) On the same diagram sketch the graph of $y = -f(x)$. [2]

(b) Let $g(x) = f(x+3)$.

(i) Find $g(-3)$

(ii) Describe **fully** the transformation that maps the graph of f to the graph of g [4]

(c) Write down in the table below the domain and the range for each function:

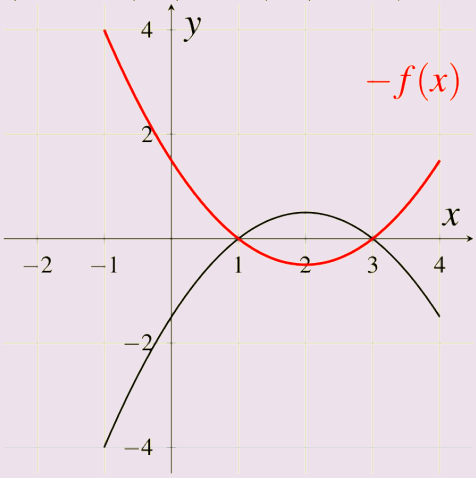
	$y = f(x)$	$y = -f(x)$	$y = f(x+3)$
Domain	$-1 \leq x \leq 4$		
Range	$-4 \leq y \leq 0.5$		

[4]

Answer 6 Source worksheet · p.13

6 *Chris · Math-AA 2.6 · Q11*

(a) Reflect in the x-axis: $y = -f(x)$ passes through $(1,0)$, $(3,0)$, min $(2,-0.5)$, $(-1,4)$, $(4,1.5)$.



(b)(i) $g(-3) = f(0) = \boxed{-1.5}$ (ii) horizontal translation 3 units to the left, vector $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$

(c) $-f(x)$: domain $\boxed{-1 \le x \le 4}$, range $\boxed{-0.5 \le y \le 4}$; $f(x+3)$: domain $\boxed{-4 \le x \le 1}$, range $\boxed{-4 \le y \le 0.5}$

Question 7

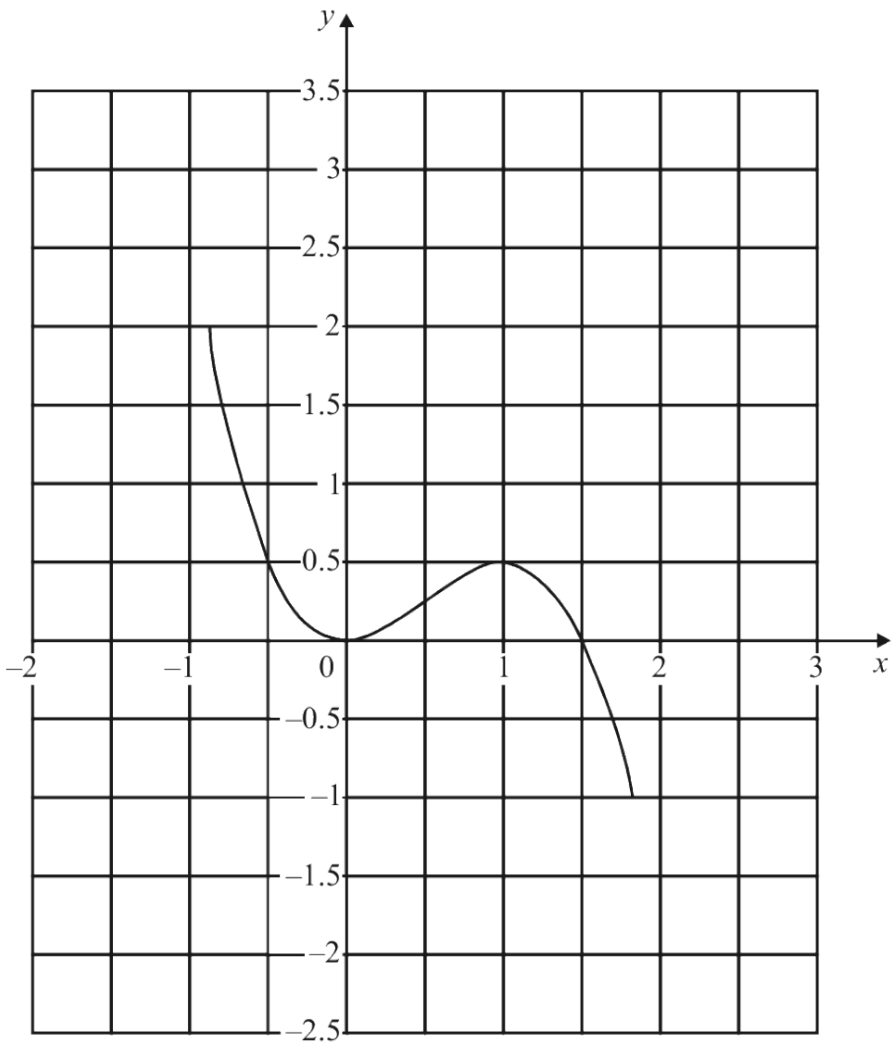
Source worksheet · p.15

7

No calculator **Chris · Math-AA 2.6 · Q12**

[Maximum mark: 4] ***[without GDC]***

The following diagram shows the graph of $y = f(x)$. It has minimum and maximum points at (0, 0) and $(1, \frac{1}{2})$.



(a) On the same diagram, draw the graph of $y = f(x-1) + \frac{3}{2}$.

(b) What are the coordinates of the minimum and maximum points of $y = f(x-1) + \frac{3}{2}$?

Answer 7

Source worksheet · p.15

7 *Chris · Math-AA 2.6 · Q12*

Every point moves 1 unit right and $\frac{3}{2}$ units up.

(a) $(-0.8, 2) \to (0.2, 3.5)$, $(0, 0) \to (1, 1.5)$, $(1, 0.5) \to (2, 2)$, $(1.8, -1) \to (2.8, 0.5)$ — draw the same shape through these points.

(b) Minimum $\boxed{\left(1, \frac{3}{2}\right)}$, maximum $\boxed{(2, 2)}$

Question 8

Source worksheet · p.17

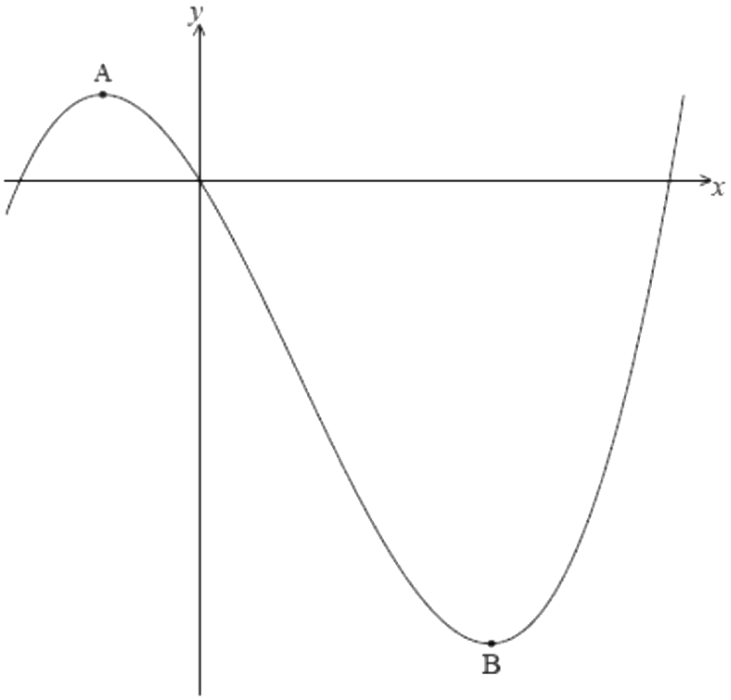
8

Calculator

Chris · Math-AA 2.6 · Q10

[Maximum mark: 8] ***[with GDC]***

Let $f(x) = \frac{1}{3}x^3 - x^2 - 3x$. Part of the graph of f is shown below.



There is a maximum point at A and a minimum point at B(3, –9).

(a) Write down the coordinates of A. [2]

(b) Write down the coordinates of

(i) the image of B after a reflection in the y-axis;

(ii) the image of B after a translation by –2 units in the x-direction and 5 units in the y -direction).

(iii) the image of B after a reflection in the x-axis followed by a horizontal stretch with scale factor $\frac{1}{2}$. [6]

Answer 8

Source worksheet · p.17

8 *Chris · Math-AA 2.6 · Q10*

(a) GDC: maximum at $\boxed{A\left(-1,\ \frac{5}{3}\right)} \approx (-1,\ 1.67)$

(b)(i) reflection in y-axis: $(x,y) \to (-x,y)$: $\boxed{(-3,\ -9)}$

(ii) $(3-2,\ -9+5) = \boxed{(1,\ -4)}$

(iii) reflection in x-axis: $(3,9)$; then horizontal stretch s.f. $\frac{1}{2}$: $\boxed{\left(\frac{3}{2},\ 9\right)}$

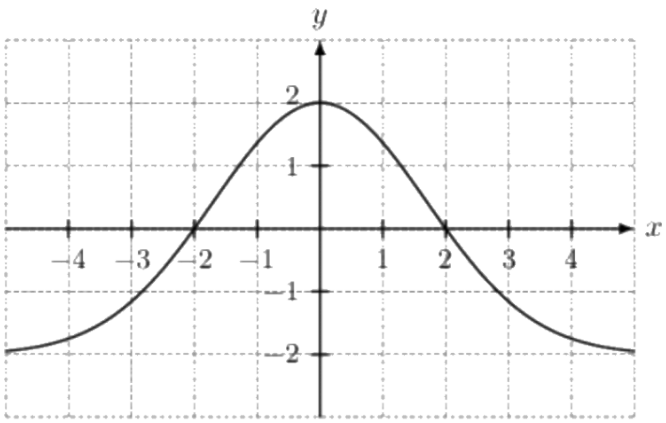
Question 9

Source worksheet · p.19

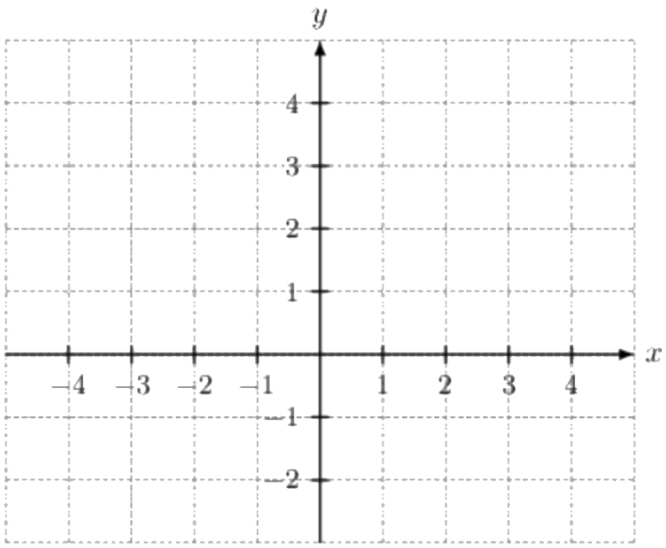
9 No calculator **RV · Revision Village Transformations · Q11**

[Maximum mark: 5]

The following diagram shows the graph of $y = f(x)$. The graph has a horizontal asymptote at $y = -2$. The graph crosses the x-axis at $x = -2$ and $x = 2$, and the y-axis at $y = 2$.



On the following set of axes, sketch the graph of $y = [f(x)]^2 - 1$, clearly showing any asymptotes with their equations and the coordinates of any local maxima or minima.



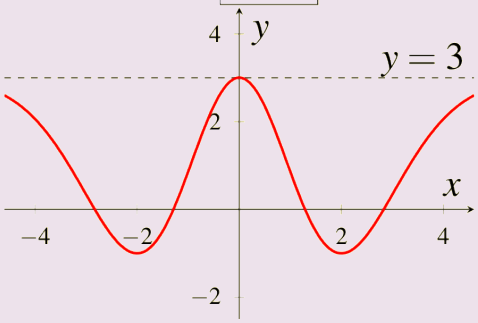
Answer 9

Source worksheet · p.19

9 *RV · Revision Village Transformations · Q11*

$y = [f(x)]^2 - 1$: at $x = 0$, $f = 2 \Rightarrow y = 3$; at $x = \pm 2$, $f = 0 \Rightarrow y = -1$; as $x \to \pm\infty$, $f \to -2 \Rightarrow y \to 4 - 1 = 3$.

Local maximum $\boxed{(0, 3)}$, local minima $\boxed{(-2, -1)}$ and $\boxed{(2, -1)}$, horizontal asymptote $\boxed{y = 3}$.



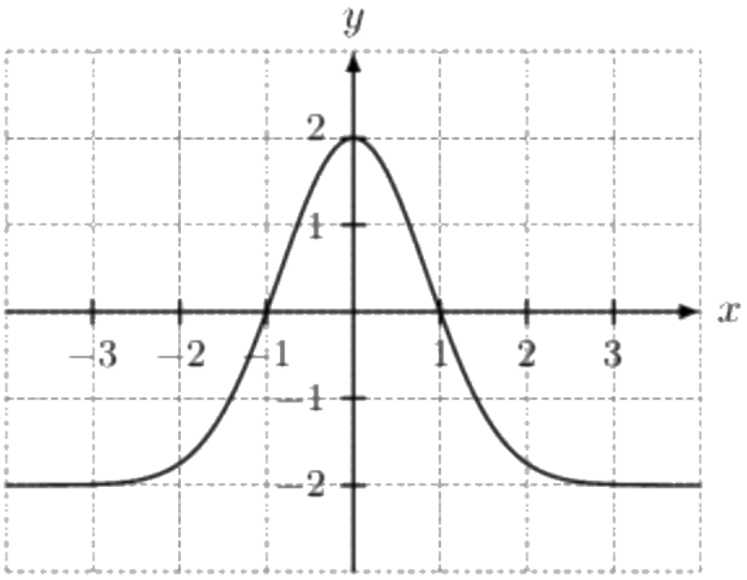
Question 10

Source worksheet · p.21

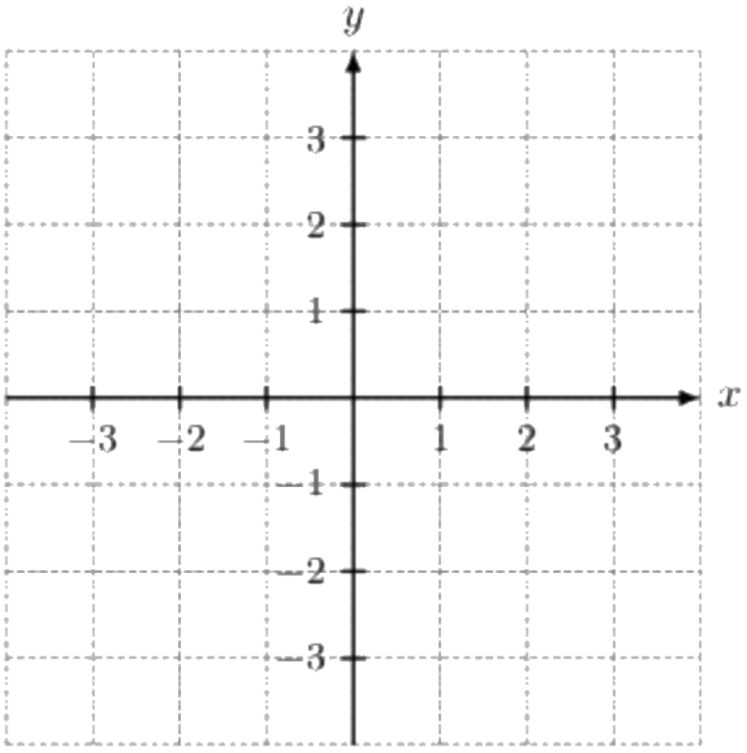
10 No calculator **RV · Revision Village Transformations · Q15**

[Maximum mark: 5]

The following diagram shows the graph of $y = f(x)$. The graph has a horizontal asymptote at $y = -2$. The graph crosses the x-axis at $x = -1$ and $x = 1$, and the y-axis at $y = 2$.



On the following set of axes, sketch the graph of $y = [f(x)]^2 - 2$, clearly showing any asymptotes with their equations and the coordinates of any local maxima or minima.



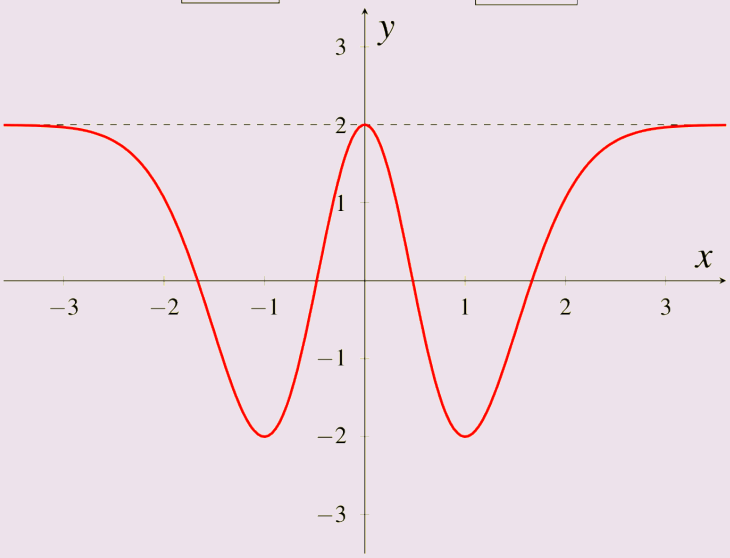
Answer 10

Source worksheet · p.21

10 *RV · Revision Village Transformations · Q15*

	$y = f(x)$	$y = [f(x)]^2$	$y = [f(x)]^2 - 2$
asymptote	$y = -2$	$y = 4$	$y = 2$
min	$(\pm 1, 0)$	$(\pm 1, 0)$	$(\pm 1, -2)$
max	$(0, 2)$	$(0, 4)$	$(0, 2)$

Asymptote $\boxed{y = 2}$; maximum $\boxed{(0, 2)}$; minima $\boxed{(-1, -2),\ (1, -2)}$

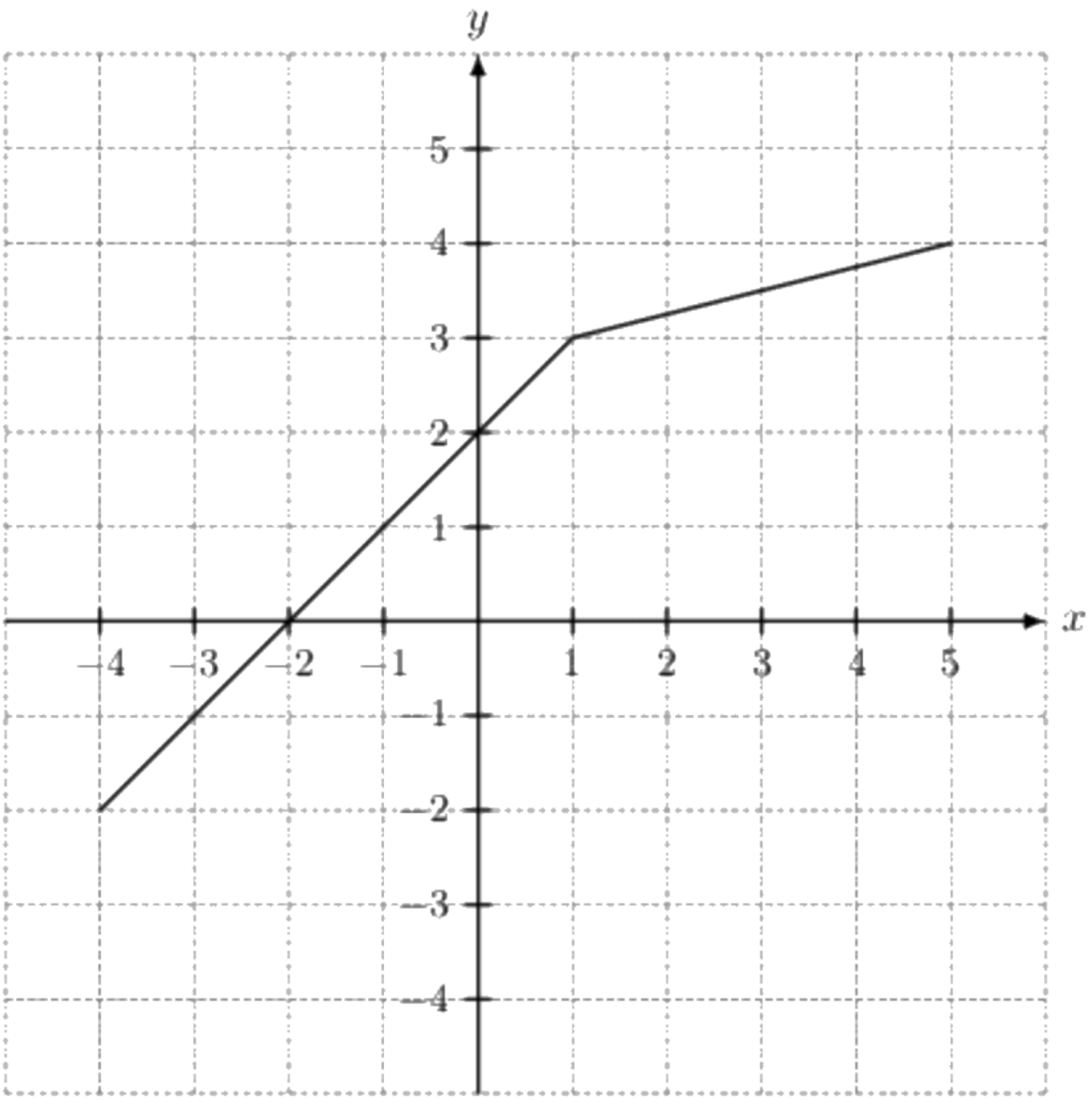


Question 11 Source worksheet · p.23

11 No calculator **RV · Rational & Inverse Functions · Q3**

[Maximum mark: 6]

The following diagram shows the graph of $y = f(x)$, for $-4 \leq x \leq 5$.



(a) Write down the value of:

(i) $f(1)$;

(ii) $f^{-1}(2)$. [2]

(b) Find the domain of f^{-1}. [2]

(c) Sketch the graph of $y = f^{-1}(x)$ on the same grid above. [2]

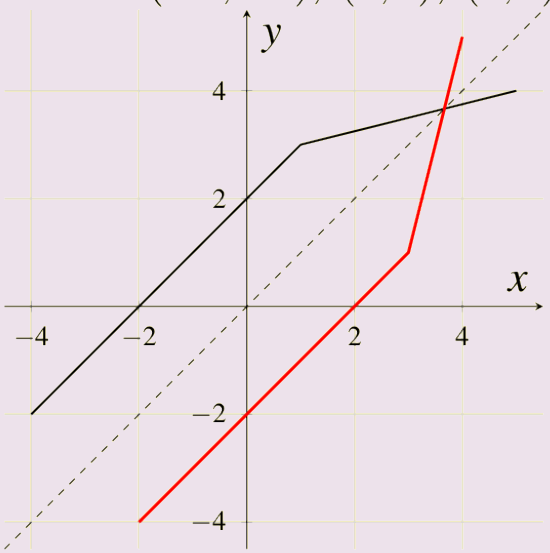
Answer 11 Source worksheet · p.23

11 *RV · Rational & Inverse Functions · Q3*

(a)(i) $\boxed{f(1)=3}$ (ii) $f(0)=2$, so $\boxed{f^{-1}(2)=0}$

(b) Domain of f^{-1} = range of f: $\boxed{-2\le x\le 4}$

(c) Reflect f in the line $y=x$: the corner points $(-4,-2),\ (1,3),\ (5,4)$ become $(-2,-4),\ (3,1),\ (4,5)$; join them with straight lines.



Question 12 Source worksheet · p.25

12 Calculator **RV · Rational & Inverse Functions · Q10**

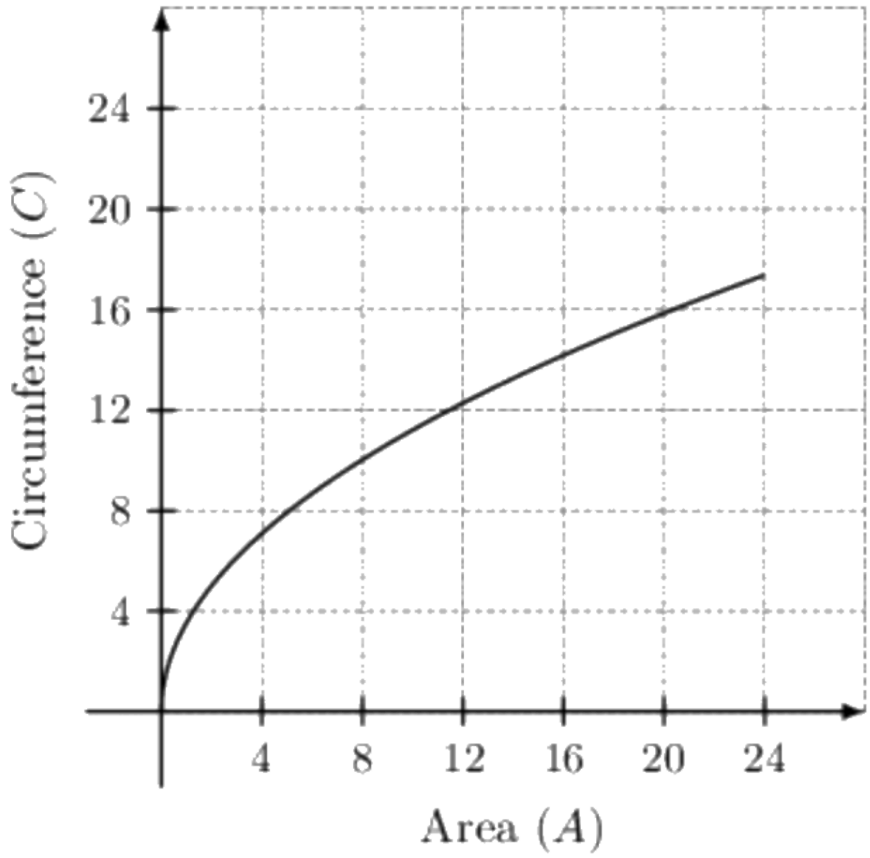
[Maximum mark: 5]

The circumference, C, of a given circle can be represented by the function

$$C(A) = 2\pi\sqrt{\frac{A}{\pi}}, \quad A \geq 0,$$

where A is the area of the circle.

The graph of the function C, for $0 \leq A \leq 24$, is shown below.



(a) Find the value of $C(24)$. [1]

(b) On the grid above, draw the graph of the inverse function, C^{-1}. [3]

(c) In the context of the question, explain the meaning of $C^{-1}(2\pi) = \pi$. [1]

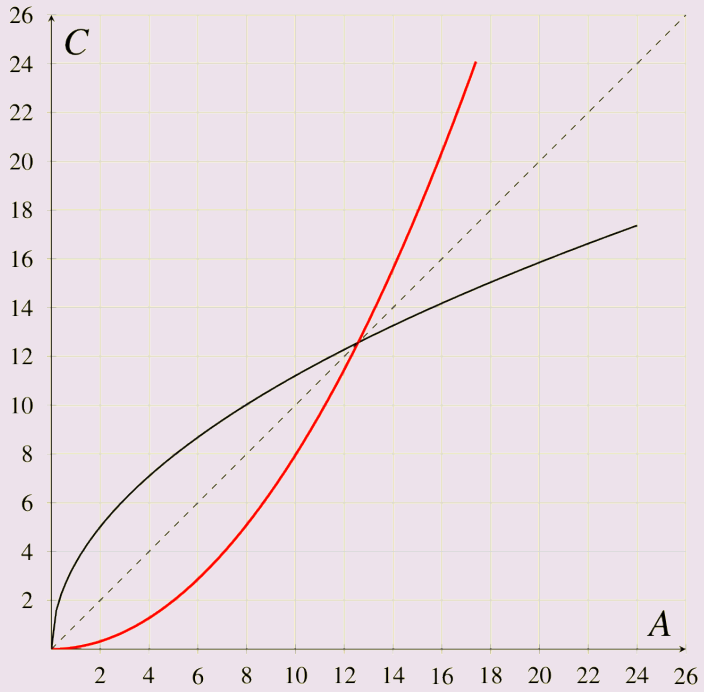
Answer 12

Source worksheet · p.25

12 *RV · Rational & Inverse Functions · Q10*

(a) $C(24) = 2\pi\sqrt{24/\pi} \approx \boxed{17.4}$

(b) Reflect the graph of C in the line $C = A$: the inverse passes through $O(0,0)$ and meets C at $(4\pi, 4\pi) \approx (12.6, 12.6)$, and is steeper beyond it ($C^{-1}(A) = \dfrac{A^2}{4\pi}$).



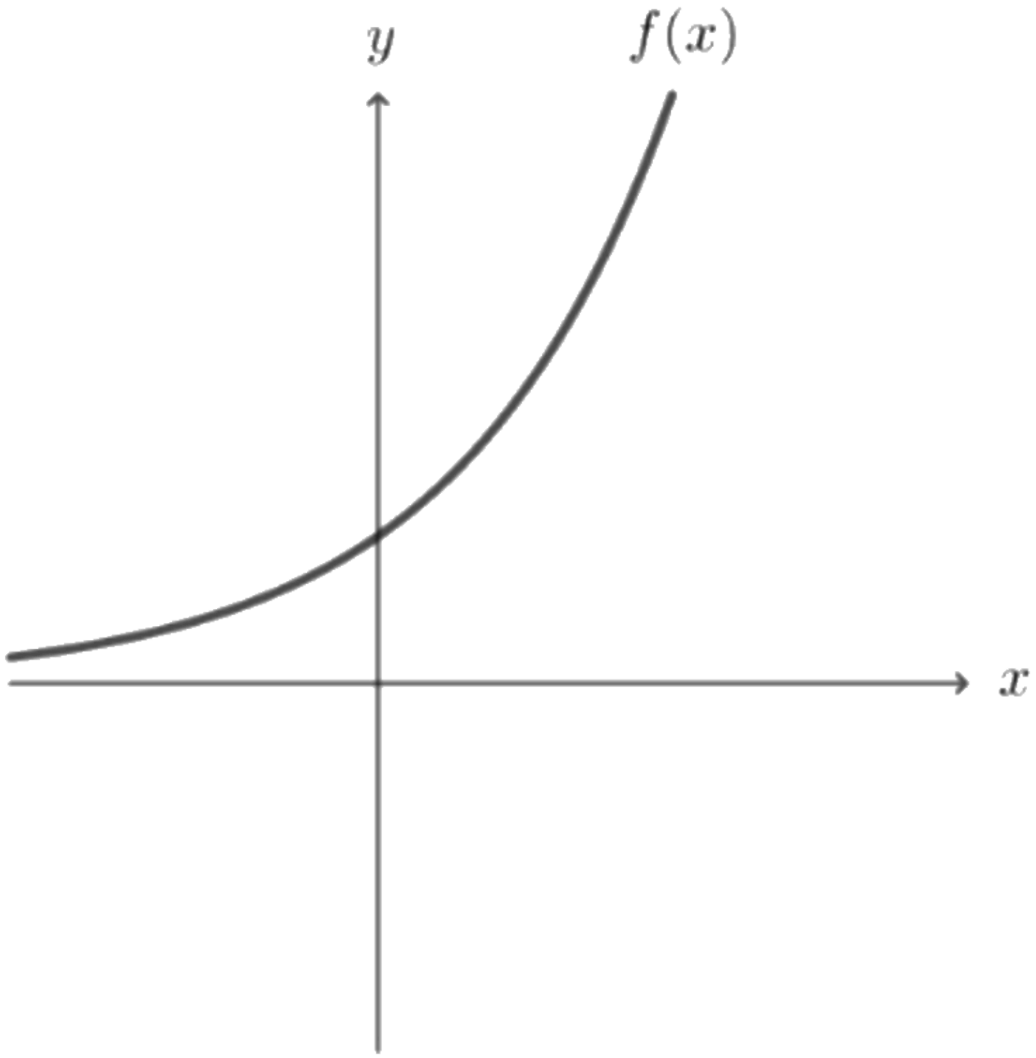
(c) $\boxed{\text{A circle whose circumference is } 2\pi \text{ has area } \pi.}$

Question 13 Source worksheet · p.27

13 No calculator **RV · Rational & Inverse Functions · Q11**

[Maximum mark: 5]

The function f is defined as $f(x) = 3e^{\frac{x}{2}}$ where $x \in \mathrm{R}$ and is shown below.



(a) Write down the equation of the asymptote of $f(x)$. [1]

(b) Write down the $y-$intercept of $f(x)$. [1]

(c) Hence, or otherwise, sketch the inverse of $f(x)$ on the image above. Make sure any asymptotes and/or intercepts are clearly marked. [2]

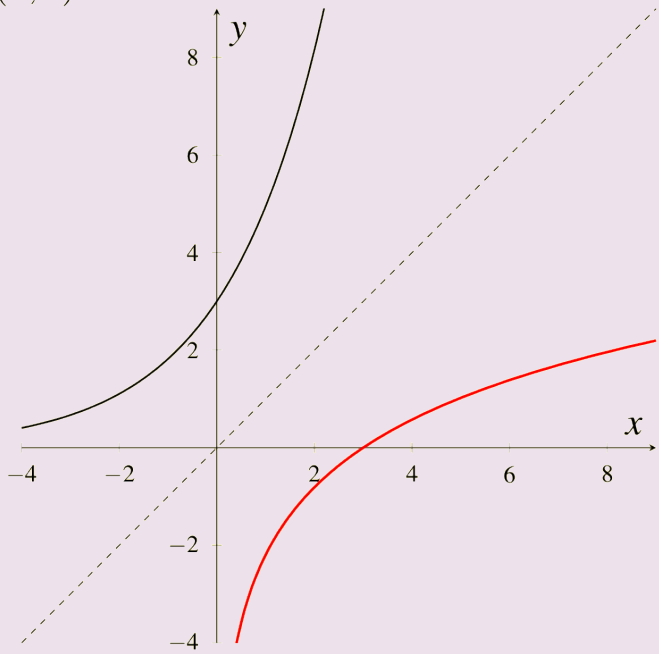
(d) Determine the domain of the inverse of $f(x)$. [1]

Answer 13 Source worksheet · p.27

13 *RV · Rational & Inverse Functions · Q11*

(a) $\boxed{y = 0}$ (b) $f(0) = 3e^0 = \boxed{3}$

(c) Reflect in $y = x$: f^{-1} has vertical asymptote $x = 0$ and x-intercept $(3, 0)$.



(d) Domain of f^{-1} = range of f: $\boxed{x > 0}$

Question 14 Source worksheet · p.29

14 No calculator **RV · Revision Village Transformations · Q2**

[Maximum mark: 6]

Let f and g be functions such that $g(x) = 3f(x-2) + 1$, for $x \in \mathbb{R}$.

The graph of g is obtained from the graph of f after the following transformations:

a vertical stretch by a factor of k, followed by
a horizontal translation of a units, followed by
a vertical translation of b units.

(a) Write down the value of:

(i) k;

(ii) a;

(iii) b. [3]

Let $h(x) = -2g(x)$, for $x \in \mathbb{R}$.

The point $\mathrm{P}(3, 4)$ on the graph of g is mapped to point Q on the graph of h.

(b) Find the coordinates of Q. [3]

Question 15 Source worksheet · p.31

15 Calculator **RV · Revision Village Transformations · Q4**

[Maximum mark: 6]

Let $f(x) = x^3$ and $g(x) = 5(x-2)^3$, for $x \in \mathbb{R}$.

The graph of g can be obtained from the graph of f using two transformations.

(a) Give a full geometric description of each of the transformations. [2]

The graph of g is translated by 4 units in the direction of the positive x-axis, and 1 unit in the direction of the negative y-axis to give the graph of a function h.

The point $\mathrm{P}(1, 1)$ on the graph of f is translated to point Q on the graph of h.

(b) Find the coordinates of Q. [4]

Answer 14 Source worksheet · p.29

14 *RV · Revision Village Transformations · Q2*

(a) $g(x) = 3f(x-2)+1$: (i) $\boxed{k=3}$ (ii) $\boxed{a=2}$ (iii) $\boxed{b=1}$

(b) $h(x) = -2g(x)$: vertical stretch s.f. 2 then reflection in x-axis.

$P(3,4) \to (3,8) \to \boxed{Q(3,-8)}$

Answer 15 Source worksheet · p.31

15 *RV · Revision Village Transformations · Q4*

(a) Vertical stretch with scale factor 5; translation 2 units in the positive x-direction.

(b) $P(1,1) \to (1,5) \to (3,5)$ on $g \to (3+4, 5-1) = \boxed{Q(7,4)}$

Question 16

Source worksheet · p.31

16 No calculator **Chris · Math-AA 2.6 · Q16**

[Maximum mark: 6] ***[without GDC]***

Let $f(x) = x^2$ and $g(x) = 2(x-1)^2$.

(a) The graph of g can be obtained from the graph of f using two transformations.

Give a full geometric description of each of the two transformations. [2]

(b) The graph of g is translated by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ (i.e. 3 units in the x-direction and –2 units in the y-direction) to give the graph of h. The point (–1, 1) on the graph of f is translated to the point P on the graph of h. Find the coordinates of P. [4]

Question 17

Source worksheet · p.33

17 No calculator **Chris · Math-AA 2.6 · Q17**

[Maximum mark: 6] ***[without GDC]***

Let $f(x) = 3x^2$. The graph of *f* is translated 1 unit to the right and 2 units down.

The graph of g is the image of the graph of f after this translation.

(a) Write down the coordinates of the vertex of the graph of g. [2]

(b) Express g in the form $g(x) = 3(x-p)^2 + q$. [2]

The graph of h is the reflection of the graph of g in the x-axis.

(c) Write down the coordinates of the vertex of the graph of h. [2]

Answer 16 Source worksheet · p.31

16 *Chris · Math-AA 2.6 · Q16*

(a) Horizontal translation 1 unit to the right, and vertical stretch with scale factor 2 (in any order).

(b) $(-1,1) \xrightarrow{g} (0,2) \xrightarrow{+\binom{3}{-2}} \boxed{P(3,0)}$ (or $h(x) = 2(x-4)^2 - 2$, $h(3) = 0$)

Answer 17 Source worksheet · p.33

17 *Chris · Math-AA 2.6 · Q17*

(a) Vertex of f is $(0,0)$, so vertex of g is $\boxed{(1,-2)}$

(b) $\boxed{g(x) = 3(x-1)^2 - 2}$ $(p = 1, q = -2)$

(c) Reflection in the x-axis: $\boxed{(1,2)}$

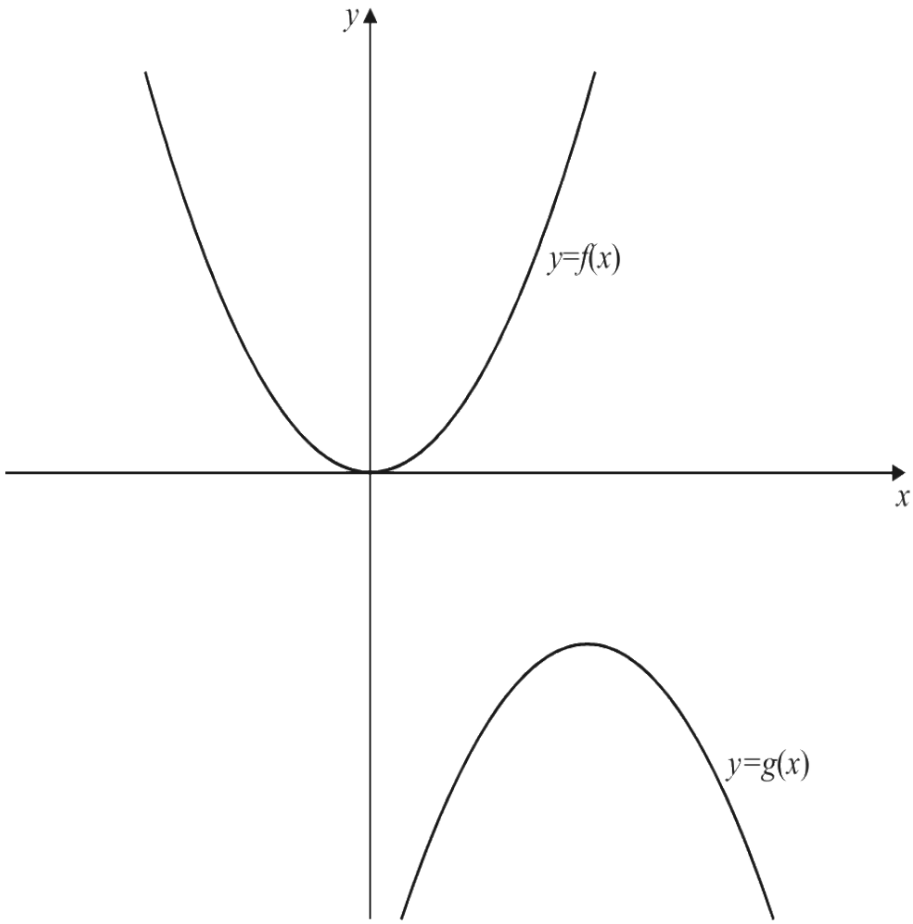
Question 18 Source worksheet · p.35

18

No calculator **Chris · Math-AA 2.6 · Q18**

[Maximum mark: 6] ***[without GDC]***

The diagram shows a sketch of part of the graph of $f(x) = x^2$ and a sketch of part of the graph of $g(x) = -x^2 + 6x - 13$.



(a) Write down the coordinates of the maximum point of $y = g(x)$

(b) Describe fully each of the transformations, which together map the graph of $y = f(x)$ onto the graph of $y = g(x)$ in a correct order,

Answer 18 Source worksheet · p.35

18 *Chris · Math-AA 2.6 · Q18*

(a) $g(x) = -(x^2 - 6x) - 13 = -(x-3)^2 + 9 - 13 = -(x-3)^2 - 4$: maximum $\boxed{(3, -4)}$

(b) Reflection in the x-axis, **followed by** a translation by the vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ (3 right, 4 down).

Question 19

Source worksheet · p.37

19 No calculator **Chris · Math-AA 2.6 · Q19**

[Maximum mark: 6] ***[with / without GDC]***

(a) Express $y = 2x^2 - 12x + 23$ in the form $y = 2(x-c)^2 + d$. [3]

The graph of $y = x^2$ is transformed into the graph of $y = 2x^2 - 12x + 23$ by the transformations

a vertical stretch with scale factor k **followed by**
a horizontal translation of p units **followed by**
a vertical translation of q units.

(b) Write down the value of (i) k; (ii) p; (iii) q. [3]

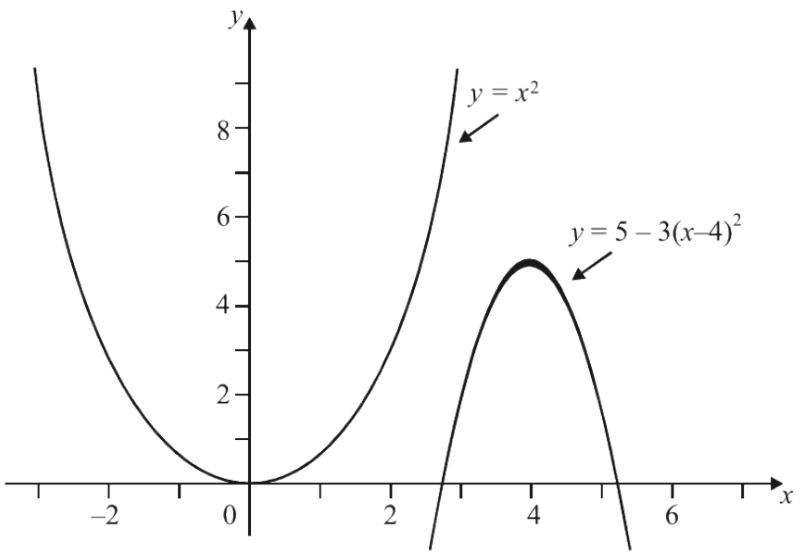
Question 20

Source worksheet · p.39

20 No calculator **Chris · Math-AA 2.6 · Q20**

[Maximum mark: 4] ***[without GDC]***

The diagram shows parts of the graphs of $y = x^2$ and $y = 5 - 3(x-4)^2$.



The graph of $y = x^2$ may be transformed into the graph of $y = 5 - 3(x-4)^2$ by these transformations.

A reflection in the line $y = 0$ **followed by**
a vertical stretch with scale factor k **followed by**
a horizontal translation of p units **followed by**
a vertical translation of q units.

Write down the value of (i) k; (ii) p; (iii) q.

Answer 19 Source worksheet · p.37

19 *Chris · Math-AA 2.6 · Q19*

(a) $2x^2 - 12x + 23 = 2(x^2 - 6x) + 23 = 2(x - 3)^2 - 18 + 23 =$

$\boxed{2(x-3)^2+5}$

(b)(i) $\boxed{k = 2}$ (ii) $\boxed{p = 3}$ (iii) $\boxed{q = 5}$

Answer 20 Source worksheet · p.39

20 *Chris · Math-AA 2.6 · Q20*

$y = x^2 \to -x^2 \to -3x^2 \to -3(x-4)^2 \to -3(x-4)^2+5$

(i) $\boxed{k = 3}$ (ii) $\boxed{p = 4}$ (iii) $\boxed{q = 5}$

Question 21 Source worksheet · p.41

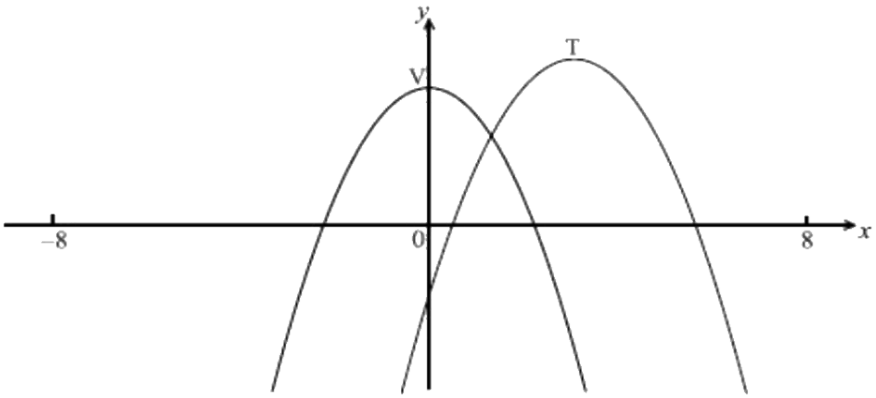
21 No calculator **Chris · Math-AA 2.6 · Q21**

[Maximum mark: 6] ***[without GDC]***

A translation with vector $\begin{pmatrix} h \\ k \end{pmatrix}$ implies a shift by h units horizontally and k units vertically.

The following diagram shows part of the graph of $f(x) = 5 - x^2$ with vertex V(0, 5).

Its image $y = g(x)$ after a translation with vector $\begin{pmatrix} h \\ k \end{pmatrix}$ has vertex T(3, 6).



(a) Write down the value of (i) h (ii) k

(b) Write down an expression for $g(x)$.

(c) On the same diagram, sketch the graph of $y = g(-x)$

Question 22 Source worksheet · p.41

22 No calculator **Chris · Math-AA 2.6 · Q24**

[Maximum mark: 6] ***[without GDC]***

The graph of $y = 2x^2 + 4x + 7$ is translated using the vector $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. Find the equation of the translated graph, giving your answer in the form $y = ax^2 + bx + c$.

Question 23 Source worksheet · p.43

23 No calculator **Chris · Math-AA 2.6 · Q25**

[Maximum mark: 6] ***[without GDC]***

The graph of the function $f(x) = 2x^3 - 3x^2 + x + 1$ is translated to its image, $g(x)$, by the vector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Write $g(x)$ in the form $g(x) = ax^3 + bx^2 + cx + d$.

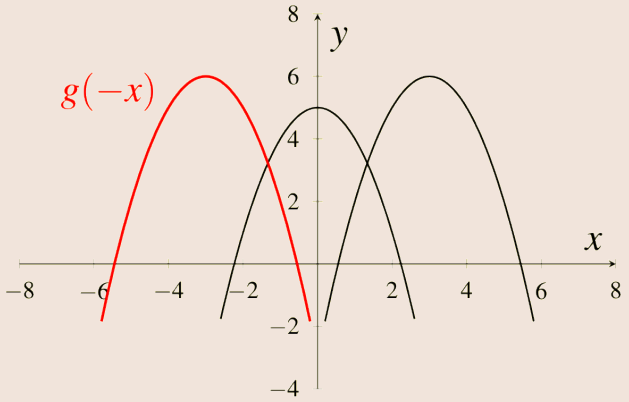
Answer 21 Source worksheet · p.41

21 *Chris · Math-AA 2.6 · Q21*

(a) $V(0,5) \to T(3,6)$: (i) $\boxed{h=3}$ (ii) $\boxed{k=1}$

(b) $g(x) = f(x-3)+1 = 5-(x-3)^2+1 = \boxed{6-(x-3)^2}$ $(= -x^2+6x-3)$

(c) $y = g(-x)$ is the reflection of g in the y-axis: vertex $(-3,6)$.



Answer 22 Source worksheet · p.41

22 *Chris · Math-AA 2.6 · Q24*

Replace x by $x-2$ and subtract 1:

$y = 2(x-2)^2+4(x-2)+7-1 = 2x^2-8x+8+4x-8+6 =$

$\boxed{2x^2-4x+6}$

Answer 23 Source worksheet · p.43

23 *Chris · Math-AA 2.6 · Q25*

$g(x) = f(x-1)-1 = 2(x-1)^3-3(x-1)^2+(x-1)+1-1$

$= 2x^3-6x^2+6x-2-3x^2+6x-3+x-1 = \boxed{2x^3-9x^2+13x-6}$

Question 24

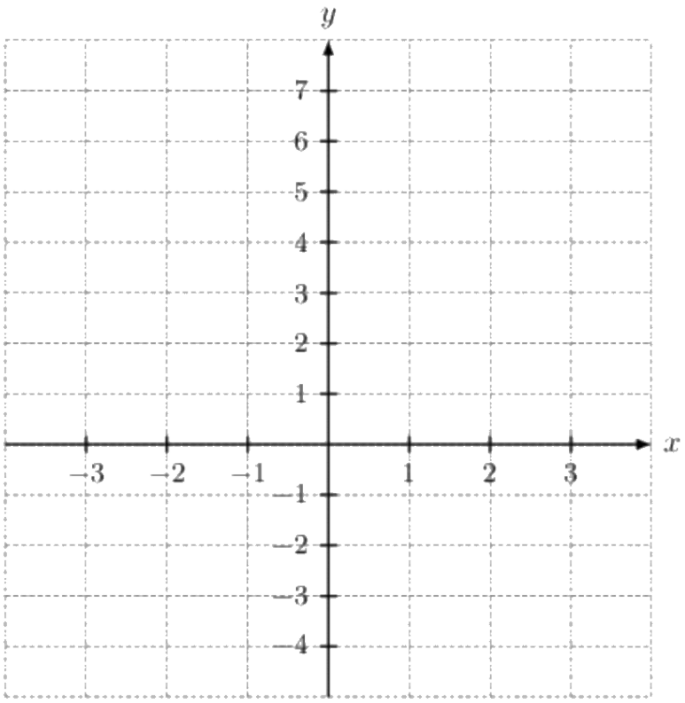
Source worksheet · p.45

24 Calculator **RV · Revision Village Transformations · Q3**

[Maximum mark: 7]

Let $f(x) = 0.2e^{x+2} - 4$, for $-3 \leq x \leq 2$.

(a) On the following grid, sketch the graph of $y = f(x)$. [3]



(b) Find the coordinates of:

(i) the x-intercept;

(ii) the y-intercept. [2]

The graph of f is reflected in the x-axis, then translated 1 unit in the direction of the positive x-axis and 2 units in the direction of the positive y-axis to obtain the graph of a function g.

(c) Find $g(x)$. [2]

Answer 24 Source worksheet · p.45

24 *RV · Revision Village Transformations · Q3*

(a) Increasing exponential from $(-3,-3.93)$ (near the asymptote $y=-4$) up to $(2,6.92)$, through $(0,-2.52)$ and $(0.996,0)$.

(b)(i) $0.2e^{x+2}=4 \Rightarrow x=\ln 20-2\approx 0.996$: $\boxed{(0.996,\,0)}$ (ii) $0.2e^{2}-4\approx -2.52$: $\boxed{(0,\,-2.52)}$

(c) $g(x)=-f(x-1)+2=-\left[0.2e^{x+1}-4\right]+2=\boxed{-0.2e^{x+1}+6}$

Question 25 Source worksheet · p.47

25 No calculator **RV · Revision Village Transformations · Q6**

[Maximum mark: 6]

The function f is defined by

$$f(x) = \begin{cases} 4x - 1 & \text{if} \quad x \geq 1, \\ -x^2 + x + 3 & \text{if} \quad x < 1. \end{cases}$$

(a) Determine whether or not f is continuous at $x = 1$. [2]

The graph of the function g is obtained by applying the following transformations to the graph of f:

a horizontal translation 2 units to the left, followed by

a reflection in the x-axis, followed by

a vertical stretch by a factor of 3.

(b) Find $g(x)$. [4]

Question 26 Source worksheet · p.47

26 No calculator **RV · Revision Village Transformations · Q13**

[Maximum mark: 5]

Consider $f(x) = 2\cos\left(x - \frac{\pi}{2}\right) + 3$ and $g(x) = 4\cos\left(x + \frac{\pi}{2}\right) + 2$.

The function f is mapped onto g by three transformations.

(a) Fully describe each of the transformations and the order in which they must be applied. [3]

A new function is such that $h(x) = g(x) + k$ where $k \in \mathbb{R}$.

(b) Find the minimum value of k such that $h(x) \geq 0$ for all x. [2]

Answer 25 Source worksheet · p.47

25 *RV · Revision Village Transformations · Q6*

(a) $f(1) = 4(1) - 1 = 3$; $\lim_{x\to 1^-} f(x) = -1 + 1 + 3 = 3$. Equal, so

$\boxed{f \text{ is continuous at } x = 1}$

(b) $f(x+2)$, then $-f(x+2)$, then $-3f(x+2)$:

$$\boxed{g(x) = \begin{cases} -12x - 21, & x \geq -1 \\ 3(x+2)^2 - 3x - 15, & x < -1 \end{cases}}$$

Answer 26 Source worksheet · p.47

26 *RV · Revision Village Transformations · Q13*

(a) $f(x+\pi) = 2\cos\left(x+\frac{\pi}{2}\right) + 3$; $2f(x+\pi) = 4\cos\left(x+\frac{\pi}{2}\right) + 6$;

$2f(x+\pi) - 4 = g(x)$.

$\boxed{\text{horizontal translation } \pi \text{ left, then vertical stretch s.f. 2, then vertical transla}}$

(The horizontal shift can be done at any point; the stretch must come before the -4.)

(b) Minimum of g is $-4+2 = -2$, so $h(x) = g(x) + k \geq 0$ needs $k \geq 2$:

$\boxed{k = 2}$

Complete transformation description (a): translate π units left, stretch vertically by scale factor 2, then translate 4 units down. The horizontal shift can occur before or after the vertical steps, but the vertical stretch must occur before the downward shift.

Question 27

Source worksheet · p.49

27 No calculator **RV · Revision Village Transformations · Q14**

[Maximum mark: 7]

Consider the functions $f(x) = 3\cos(x) + \frac{9}{2}$ and $g(x) = 3\cos\left(x + \frac{\pi}{3}\right) + A$, where $x \in \mathbb{R}$ and $A < \frac{9}{2}$.

(a) Describe a sequence of two transformations that transforms the graph of f to the graph of g. [3]

The y-intercept of the graph g is at the point $\left(0, \frac{9}{2}\right)$

(b) Find the range of g. [4]

Answer 27

Source worksheet · p.49

27 *RV · Revision Village Transformations · Q14*

(a) $f\left(x+\frac{\pi}{3}\right)=3\cos\left(x+\frac{\pi}{3}\right)+\frac{9}{2}$, then subtract $\frac{9}{2}-A$.

translation $\frac{\pi}{3}$ in the negative x-direction; translation $\frac{9}{2}-A$ in the negative y-

(any order)

(b) $g(0)=3\cos\frac{\pi}{3}+A=\frac{3}{2}+A=\frac{9}{2}\Rightarrow A=3.$ $g(x)=3\cos\left(x+\frac{\pi}{3}\right)+3$:

range $0\leq g(x)\leq 6$

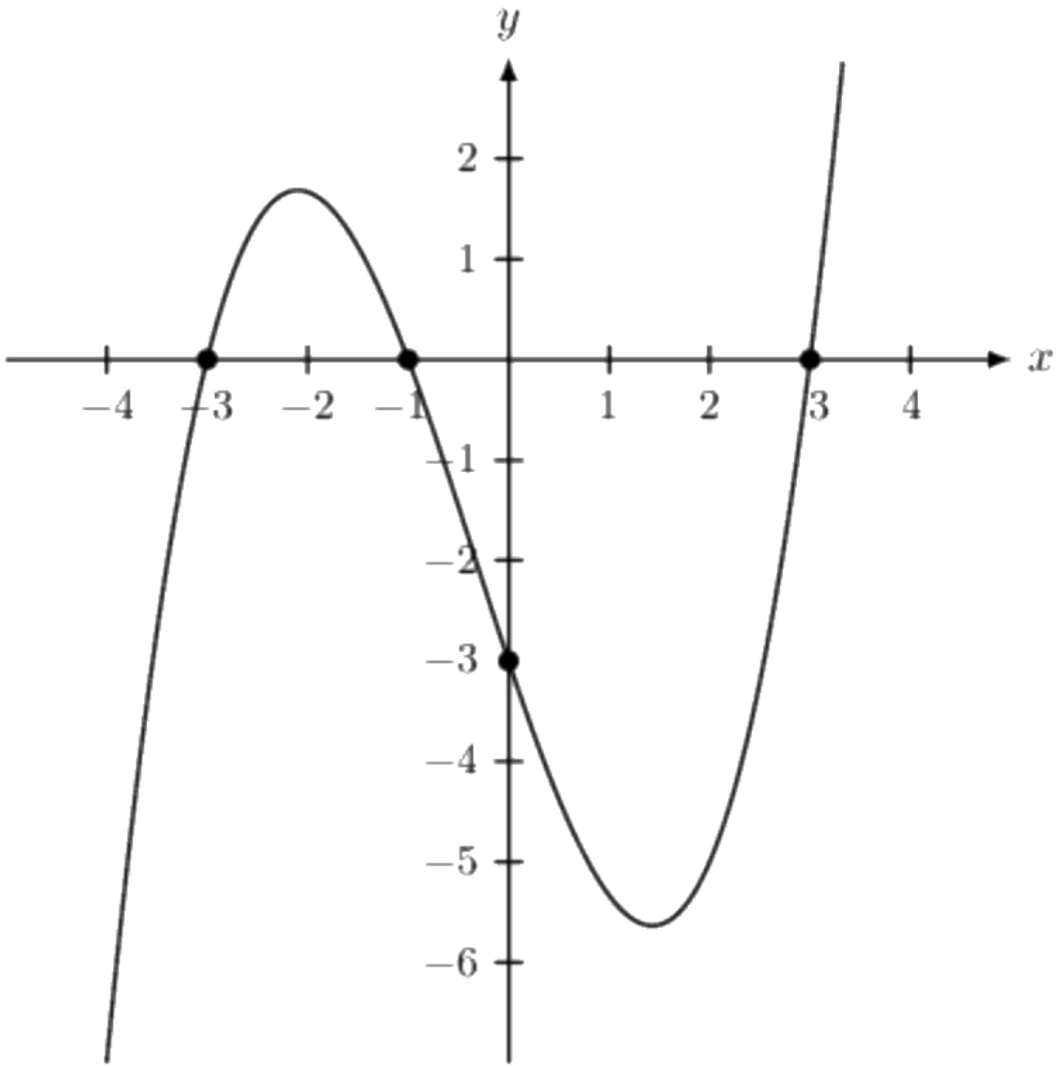
Complete transformation description (a): translate $\frac{\pi}{3}$ units left and then translate vertically by $A-\frac{9}{2}$. With the value $A=3$ found in (b), this is $\frac{3}{2}$ units down.

Question 28 Source worksheet · p.51

28 No calculator **RV · Revision Village Transformations · Q17**

[Maximum mark: 7]

Let $f(x) = ax^3 + bx^2 + cx + d$, for $x \in \mathbb{R}$, where $a, b, c, d \in \mathbb{Q}$. The diagram below shows part of the graph of $y = f(x)$.



(a) Using the information shown in the diagram, find the values of a, b, c and d. [4]

(b) Let $g(x) = -\frac{3}{4}f(-2x + 1)$.

(i) Find the coordinates of the points where the graph of $y = g(x)$ intercepts the x-axis.

(ii) Find the y-intercept of the graph of $y = g(x)$. [3]

Answer 28

Source worksheet · p.51

28 *RV · Revision Village Transformations · Q17*

(a) Roots $-3,-1,3$: $f(x)=a(x+3)(x+1)(x-3)=a(x^3+x^2-9x-9)$;

$f(0)=-9a=-3\Rightarrow a=\frac{1}{3}$

$f(x)=\frac{1}{3}x^3+\frac{1}{3}x^2-3x-3$: $\boxed{a=b=\frac{1}{3},\ c=d=-3}$

(b)(i) $f(-2x+1)=0\Rightarrow -2x+1=-3,-1,3\Rightarrow x=2,1,-1$:

$\boxed{(-1,0),\ (1,0),\ (2,0)}$

(ii) $g(0)=-\frac{3}{4}f(1)=-\frac{3}{4}\left(\frac{1}{3}+\frac{1}{3}-3-3\right)=-\frac{3}{4}\left(-\frac{16}{3}\right)=4$: $\boxed{(0,4)}$

Question 29

Source worksheet · p.53

29

No calculator

Chris · Math-AA 2.6 · Q22

[Maximum mark: 6] ***[with / without GDC]***

The quadratic function f is defined by $f(x) = 3x^2 - 12x + 11$.

(a) Write f in the form $f(x) = 3(x-h)^2 + k$.

(b) The graph of f is translated 3 units in the positive x-direction and 5 units in the positive y-direction. Find the function g for the translated graph, giving your answer in the form $g(x) = 3(x-p)^2 + q$.

Answer 29

Source worksheet · p.53

29 *Chris · Math-AA 2.6 · Q22*

(a) $3x^2 - 12x + 11 = 3(x^2 - 4x) + 11 = 3(x - 2)^2 - 12 + 11 =$

$\boxed{3(x-2)^2 - 1}$

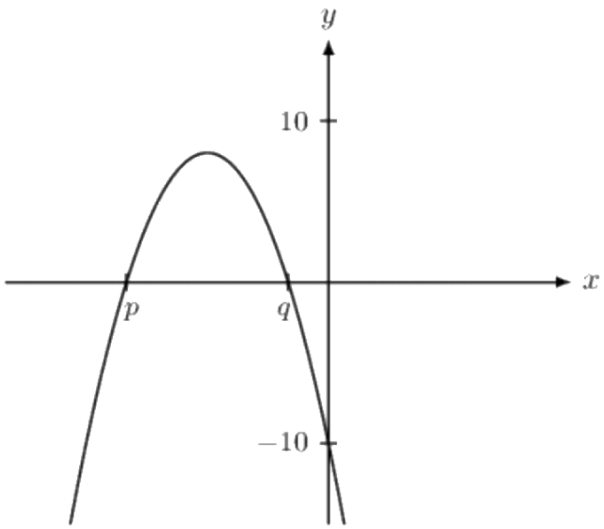
(b) Vertex $(2, -1) \to (5, 4)$: $\boxed{g(x) = 3(x-5)^2 + 4}$

Question 30 Source worksheet · p.55

30 No calculator **RV · Revision Village Transformations · Q7**

[Maximum mark: 15]

Let $f(x) = a(x+1)(x+5)$, for $x \in \mathbb{R}$, where $a \in \mathbb{Z}$. The following diagram shows part of the graph of f.



The graph of f has x-intercepts at $(p, 0)$ and $(q, 0)$, and a y-intercept at $(0, -10)$.

(a) (i) Write down the value of p and the value of q.

(ii) Find the value of a. [6]

(b) Find the equation of the axis of symmetry. [2]

(c) Find the coordinates of the vertex. [3]

The graph of a function g is obtained from the graph of f by a reflection in the y-axis, followed by a translation of 2 units in the direction of the positive y-axis. The point $\mathrm{P}(-2, 6)$ on the graph of f is mapped to point Q on the graph of g.

(d) Find the coordinates of Q. [4]

Answer 30

Source worksheet · p.55

30 *RV · Revision Village Transformations · Q7*

(a)(i) $\boxed{p = -5,\ q = -1}$ (ii) $f(0) = 5a = -10 \Rightarrow \boxed{a = -2}$

(b) $f(x) = -2x^2 - 12x - 10,\ x = -\dfrac{b}{2a} = -\dfrac{-12}{-4}$: $\boxed{x = -3}$

(c) $f(-3) = -2(-2)(2) = 8$: $\boxed{(-3, 8)}$

(d) Reflect in y-axis then 2 up: $P(-2,6) \to (2,6) \to \boxed{Q(2, 8)}$

Question 31

Source worksheet · p.57

31 No calculator **RV · Revision Village Transformations · Q8**

[Maximum mark: 15]

Let $f(x) = 3x^2 + 12x + 9$, for $x \in \mathbb{R}$.

(a) For the graph of f, find:

(i) the y-intercept;

(ii) the x-intercepts. [4]

The function f can be written in the form $f(x) = a(x - h)^2 + k$.

(b) Find the values of a, h and k. [5]

(c) For the graph of f, write down:

(i) the coordinates of the vertex;

(ii) the equation of the axis of symmetry. [2]

The graph of a function g is obtained from the graph of f by a reflection in the x-axis, followed by a translation of 4 units in the direction of the positive y axis.

(d) Find $g(x)$, giving your answer in the form $g(x) = px^2 + qx + r$. [4]

Answer 31

Source worksheet · p.57

31 *RV · Revision Village Transformations · Q8*

(a)(i) $\boxed{(0, 9)}$ (ii) $3(x+3)(x+1) = 0$: $\boxed{(-3,0),\ (-1,0)}$

(b) $3(x^2+4x)+9 = 3(x+2)^2-12+9 = 3(x+2)^2-3$: $\boxed{a = 3,\ h = -2,\ k =}$

(c)(i) $\boxed{(-2, -3)}$ (ii) $\boxed{x = -2}$

(d) $g(x) = -f(x)+4 = -3x^2-12x-9+4 = \boxed{-3x^2-12x-5}$

Question 32

Source worksheet · p.59

32 No calculator **RV · Revision Village Transformations · Q9**

[Maximum mark: 15]

Let $f(x) = 2x^2 + 4x + p$, for $x \in \mathbb{R}$, where $p \in \mathbb{Z}$.

(a) The equation $f(x) = 0$ has two equal roots.

(i) Write down the value of the discriminant of f.

(ii) Show that $p = 2$. [3]

(b) For the graph of f, find:

(i) the equation of the axis of symmetry.

(ii) the coordinates of the vertex; [4]

(c) Write down the solution to the equation $f(x) = 0$. [1]

(d) The function f can be written in the form $f(x) = a(x - h)^2 + k$. Find the values of a, h and k. [5]

(e) The graph of a function g is obtained from the graph of f by a reflection in the x-axis. Find the coordinates of the vertex of the graph of g. [2]

Answer 32 Source worksheet · p.59

32 *RV · Revision Village Transformations · Q9*

(a)(i) $\boxed{\Delta = 0}$ (ii) $\Delta = 4^2 - 4(2)(p) = 16 - 8p = 0 \Rightarrow p = 2$ ✓

(b)(i) $x = -\dfrac{4}{2(2)}$: $\boxed{x = -1}$ (ii) $f(-1) = 2 - 4 + 2 = 0$: $\boxed{(-1, 0)}$

(c) $\boxed{x = -1}$

(d) $2(x^2+2x)+2 = 2\left[(x+1)^2 - 1\right] + 2 = 2(x+1)^2$: $\boxed{a = 2,\ h = -1,\ k = 0}$

(e) $g(x) = -f(x) = -2(x+1)^2$: vertex $\boxed{(-1, 0)}$ (on the x-axis, so it does not move)

Question 33

Source worksheet · p.61

33 No calculator **RV · Revision Village Transformations · Q10**

[Maximum mark: 15]

Let $f(x) = 2x^2 - 8x + 6$, for $x \in \mathbb{R}$.

(a) Write down the value of $f(0)$. [1]

(b) Solve the equation $f(x) = 0$. [3]

The function f can be written in the form $f(x) = a(x - h)^2 + k$.

(c) Find the values of a, h and k. [5]

(d) For the graph of f, write down:

(i) the coordinates of the vertex;

(ii) the equation of the axis of symmetry. [2]

The graph of a function g is obtained from the graph of f by a reflection in the x-axis, followed by a translation of 1 unit to the right and 3 units up.

(e) Find $g(x)$, giving your answer in the form $g(x) = px^2 + qx + r$. [4]

Answer 33

Source worksheet · p.61

33 *RV · Revision Village Transformations · Q10*

(a) $\boxed{f(0) = 6}$ (b) $2(x-1)(x-3) = 0$: $\boxed{x = 1,\ 3}$

(c) $2(x^2 - 4x) + 6 = 2(x-2)^2 - 2$: $\boxed{a = 2,\ h = 2,\ k = -2}$ (d)(i) $\boxed{(2, -2)}$ (ii) $\boxed{x = 2}$

(e) $g(x) = -f(x-1) + 3 = -[2(x-3)^2 - 2] + 3 = -2(x-3)^2 + 5 = \boxed{-2x^2 + 12x - 13}$

Question 34

Source worksheet · p.63

34 No calculator **RV · Revision Village Transformations · Q12**

[Maximum mark: 15]

Let $f(x) = 2(x-1)^2 - 8$, for $x \in \mathbb{R}$.

(a) Show that $f(x) = 2x^2 - 4x - 6$. [2]

(b) For the graph of f:

(i) write down the coordinates of the vertex;

(ii) write down the y-intercept;

(iii) find both x-intercepts. [7]

(c) Hence sketch the graph of f. [3]

Let $g(x) = 6x^2$, for $x \in \mathbb{R}$.

The graph of f may be obtained from the graph of g using the following three transformations:

- a compression of scale factor a in the y-direction, followed by
- a horizontal translation of h units, followed by
- a vertical translation of k units.

(d) Find the values of a, h and k. [3]

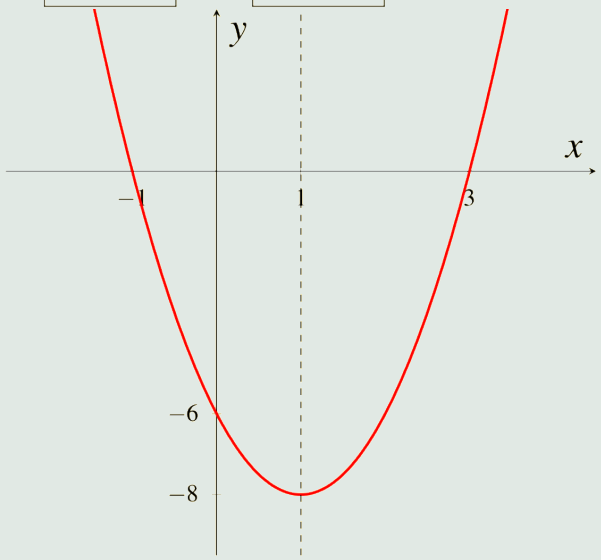
Answer 34 Source worksheet · p.63

34 *RV · Revision Village Transformations · Q12*

(a) $2(x^2-2x+1)-8=2x^2-4x-6$ ✓

(b)(i) $\boxed{(1,-8)}$ (ii) $\boxed{(0,-6)}$ (iii) $2(x+1)(x-3)=0$: $\boxed{(-1,0),\ (3,0)}$

(c)



(d) $f(x)=a\,g(x-h)+k=6a(x-h)^2+k=2(x-1)^2-8$: $\boxed{a=\frac{1}{3},\ h=1,\ k=}$

Question 35

Source worksheet · p.65

35 Calculator **Chris · Math-AA 2.6 · Q26**

[Maximum mark: 15] ***[with GDC]***

Let $f(x) = 3(x+1)^2 - 12$.

(a) Show that $f(x) = 3x^2 + 6x - 9$. [2]

(b) For the graph of f

(i) write down the coordinates of the vertex;

(ii) write down the **equation** of the axis of symmetry;

(iii) write down the y-intercept;

(iv) find both x-intercepts. [8]

(c) **Hence** sketch the graph of f. [2]

(d) Let $g(x) = x^2$. The graph of f may be obtained from the graph of g by the two transformations:

a stretch of scale factor t in the y-direction

followed by a translation of $\begin{pmatrix} p \\ q \end{pmatrix}$.

Find $\begin{pmatrix} p \\ q \end{pmatrix}$ and the value of t. [3]

Answer 35 Source worksheet · p.65

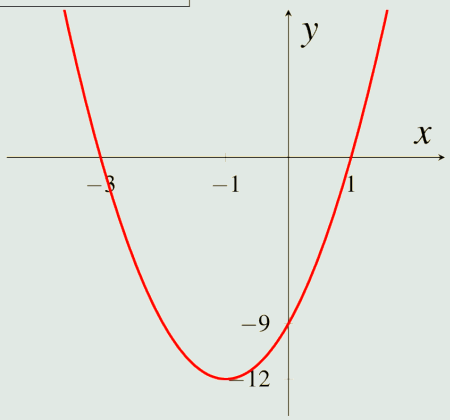
35 *Chris · Math-AA 2.6 · Q26*

(a) $3(x^2+2x+1)-12 = 3x^2+6x-9$ ✓

(b)(i) $\boxed{(-1,-12)}$ (ii) $\boxed{x=-1}$ (iii) $\boxed{(0,-9)}$ (iv) $3(x+3)(x-1)=0$:

$\boxed{(-3,0),\ (1,0)}$

(c)



(d) Stretch s.f. 3 in y: $3x^2$; translate: $3(x+1)^2-12$. $\boxed{\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} -1 \\ -12 \end{pmatrix},\ t =$

Question 36 Source worksheet · p.67

36 No calculator **Practice · Rational functions · Q1**

Let $f(x) = \frac{1}{x}$ and $g(x) = \frac{3}{x-2} + 1$.

(a) Describe fully a sequence of transformations that maps the graph of f onto the graph of g. [3]

(b) Write down the equations of the asymptotes of the graph of g. [2]

(c) Write down the domain and range of g. [2]

Question 37 Source worksheet · p.67

37 No calculator **Practice · Rational functions · Q2**

Let $f(x) = \frac{2x+5}{x+1}$, $x \neq -1$.

(a) Show that $f(x) = 2 + \frac{3}{x+1}$. [2]

(b) Hence describe the transformations that map the graph of $y = \frac{1}{x}$ onto the graph of $y = f(x)$. [3]

(c) Write down the equations of the asymptotes of the graph of f. [2]

Question 38 Source worksheet · p.69

38 No calculator **Practice · Rational functions · Q3**

The graph of $y = \frac{1}{x}$ is translated by the vector $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$ and then reflected in the x-axis.

(a) Find the equation of the resulting graph. [3]

(b) Write down the equations of its asymptotes. [2]

Answer 36 Source worksheet · p.67

36 *Practice · Rational functions · Q1*

(a) Vertical stretch with scale factor 3, then translation by $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ (2 right, 1 up).

(b) $\boxed{x = 2}$ and $\boxed{y = 1}$ (c) domain $\boxed{x \in \mathbb{R},\ x \neq 2}$, range $\boxed{y \in \mathbb{R},\ y \neq 1}$

Answer 37 Source worksheet · p.67

37 *Practice · Rational functions · Q2*

(a) $\dfrac{2x+5}{x+1} = \dfrac{2(x+1)+3}{x+1} = 2 + \dfrac{3}{x+1}$ ✓

(b) Vertical stretch with scale factor 3, then translation by $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ (1 left, 2 up).

(c) $\boxed{x = -1}$ and $\boxed{y = 2}$

Answer 38 Source worksheet · p.69

38 *Practice · Rational functions · Q3*

(a) Translate: $y = \dfrac{1}{x+3} + 4$; reflect in x-axis: $\boxed{y = -\dfrac{1}{x+3} - 4}$

(b) $\boxed{x = -3}$ and $\boxed{y = -4}$